program 10

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10. Develop an R function to draw a density curve representing the probability function of a continuous variable, with separate curves for each group, using ggplot2

Step 1: Load the library

```
library(ggplot2)
```

Step 2: Define the function

```
plot_density_by_group <- function(data, numeric_var, group_var, fill = TRUE) {
  ggplot(data, aes_string(x = numeric_var, color = group_var, fill = group_var)) +
    geom_density(alpha = if (fill) 0.4 else 0) +
    labs(
      title = paste("Density Plot of", numeric_var, "by", group_var),
      x = numeric_var,
      y = "Density"
    ) +
    theme_minimal()
}
```

Step 3:Load Dataset

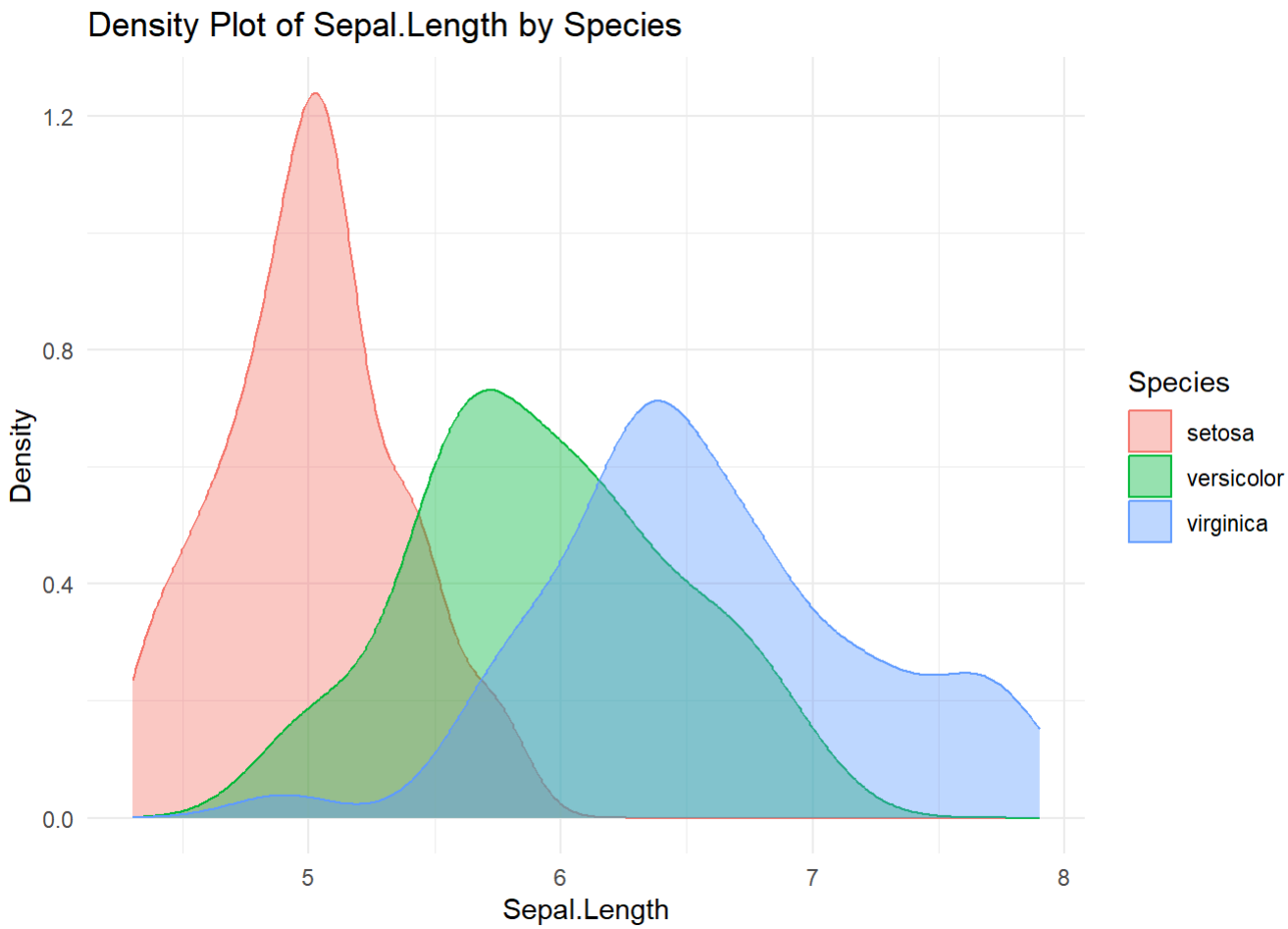
```
data(iris)
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

step 4: Call the Function with Example

```
plot_density_by_group(iris, "Sepal.Length", "Species")
```

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
ℹ Please use tidy evaluation idioms with `aes()`.
ℹ See also `vignette("ggplot2-in-packages")` for more information.
```


Density Plot of Sepal.Length by Species
1.2
0.8
0.4
0.0
Density
5
6
7
8
Sepal.Length
Species
setosa
versicolor
virginica